

PRODUCT PORTFOLIO

My AI Product Portfolio

Turning customer pain into shipped AI products — problem framing, product vision, and the judgment of when (and when not) to reach for AI.

AI Defence Ecosystem

Agentic Platform Extensions

Applied AI in Finance, Trade & Health

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8 products · **3** arenas

WHAT THIS PORTFOLIO DEMONSTRATES

From customer pain to shipped AI product



Frame the real pain

Start from the customer's actual blocker — not the technology.



Define the vision & fit

Shape solutions with clear strategic, user and business value.



Judge when to use AI

LLM, classic ML, or neither — matched to the problem, not the hype.



Architect end-to-end

Agents, RAG, microservices, governance and trust layers.



Ship tangible demos

Every product has a working, zero-install way to see it run.



Defend the business case

Value modelled honestly, with metrics flagged as targets.

The Portfolio at a Glance

Eight products across three arenas — unified by one discipline: putting AI only where it genuinely creates value, behind the right guardrails.

AI DEFENCE ECOSYSTEM

Prevent · Protect · Detect & Respond



AI Policy Foundry

Prevent — autonomous cloud-security policy from live intel + your risk register



AI-SPM

Protect — posture management for your own AI/ML assets & compliance



AI SOC Portal

Detect & Respond — auditable, AI-native SOC triage and hunting

AGENTIC PLATFORM EXTENSIONS

AI that safely drives the platform



Exploit Validation MCP

MCP server — governed validate → approve → remediate loop for AI agents



Unified Risk Copilot

Conversational risk triage across 8 modules, fully governed & audited

APPLIED AI — BEYOND SECURITY

Breadth of domain & the right model choice



HS Code Prediction API

Customs classification via classic ML — right-sized, not an LLM



KYCas-a-Service

Verifiable, privacy-preserving credit scoring — AI on blockchain



Ellyra Pulse

Safety-gated patient sentiment for a medical AI — PHI-safe, clinical-trust first

AI Policy Foundry

Autonomous cloud-security policy generation, grounded in your own risk register

Implemented for: **British Petroleum**

 [View on GitHub `github.com/mayukhg/ai-policy-foundry`](https://github.com/mayukhg/ai-policy-foundry)

BUILT WITH

React + TypeScript · Node/Express · GPT-4 multi-agent ·
PostgreSQL · Redis · WebSocket



CUSTOMER PAIN

Cloud security policy is written by hand — 2–6 weeks per policy, 60–75% coverage. Providers ship new services faster than policy keeps up, so gaps (and risk) accumulate.



THE PRODUCT BET

A multi-agent system that generates, validates and continuously updates policy from live threat intel AND the organisation's own uploaded risk/control docs — audit-ready in minutes, not weeks.



WHY AI IS THE RIGHT CALL

Input is unstructured (threat feeds, regulations, risk registers) and the mapping is non-deterministic — a language + reasoning task templating can't do. Context grounding lifts accuracy from ~65% to ~95%.



KEY TRADEOFF

Deprioritised multi-cloud breadth to first nail per-clause traceability — an auditor won't trust a policy they can't trace back to a specific control.



VALIDATED WITH

Ran the pain with BP's cloud-security policy-management team. Their weeks-long manual cycles and creeping coverage gaps are what drove risk-register grounding + continuous auto-update.



VALUE & OUTCOMES

- ~70% faster, ~94% coverage — modelled on the public build; production figures under NDA
- Fewer policy gaps → lower breach exposure
- Roadmap: threat correlation, batch docs, OCR

AI SOC Portal

An AI-native SOC workspace where every machine verdict can be traced, evaluated and audited

Implemented for: **British Petroleum**

 [View on GitHub](https://github.com/mayukhg/ai-soc-portal) github.com/mayukhg/ai-soc-portal

BUILT WITH

React + TypeScript · AWS Lambda (Python) · LangChain /
LangGraph · Pinecone · LangSmith + RAGAS



CUSTOMER PAIN

Most "AI for SOC" just wires an LLM to the alert queue. But a Tier-1 analyst won't act on a verdict they can't interrogate, and a CISO won't sign off on a control they can't evidence. Trust — not capability — blocks adoption.



THE PRODUCT BET

Agentic triage, threat hunting and incident response over live telemetry, built so every conclusion is replayable. ~Half the repo is reasoning, evaluation and audit infrastructure — trust as a product requirement.



WHY AI IS THE RIGHT CALL

Yes for triage, correlation and natural-language hunting (LLM + RAG excel here) — but only when paired with an explainability, evaluation and audit layer. Raw AI alone is a non-starter in a regulated SOC.



KEY TRADEOFF

Chose to spend the build on trust infrastructure over more raw detection features — in a regulated SOC, trust, not capability, is what gates adoption.



VALIDATED WITH

Multiple interview rounds across all levels of BP's SOC team. Hearing that analysts won't act on a verdict they can't interrogate is why ~half the build is the eval, explainability and audit layer.



VALUE & OUTCOMES

- Unlocks AI exactly where trust gates adoption
- Faster triage; defensible, replayable verdicts
- Roadmap: publish RAGAS baselines; SIEM/EDR ingest

AI Security Posture Management

Govern and secure your own AI/ML assets across their entire lifecycle

Implemented for: **British Petroleum**



View on GitHub github.com/mayukhg/ai-spm

BUILT WITH

React · Node API gateway · Python FastAPI microservices ·
Istio service mesh (mTLS) · PostgreSQL



CUSTOMER PAIN

Enterprises ship AI into production, but models, datasets, pipelines and endpoints are an unmanaged attack surface — no inventory, no adversarial detection, no drift monitoring — under rising NIST AI RMF / EU AI Act / GDPR pressure.



THE PRODUCT BET

Posture management for AI itself: inventory + lineage, adversarial-attack detection, data-quality & drift monitoring, and automated compliance evidence — a single governed view for analyst, ML engineer, compliance and CISO.



WHY AI IS THE RIGHT CALL

Deliberately applied ML/statistics, not an LLM: adversarial (FGSM/PGD/C&W), drift (KS, Chi-sq, Wasserstein), anomaly (Isolation Forest/LOF/DBSCAN). The problems are statistical, so the tooling is — judgment over hype.



KEY TRADEOFF

Chose applied ML/statistics over an LLM — the problems (adversarial, drift, anomaly) are statistical, so an LLM would add cost and risk, not accuracy.



VALIDATED WITH

Interviewed BP's cloud-security team on how they plan to secure the org's AI initiatives. The absence of any AI-asset inventory or governance is what scoped this to posture management for AI itself.



VALUE & OUTCOMES

- Makes AI adoption governable & auditable
- Automated compliance evidence, Wiz-native
- Roadmap: broader frameworks & integrations

SPOTLIGHT — THE PLATFORM PLAY

The AI Defence Ecosystem

Three products, one platform, the full lifecycle — because AI both defends the enterprise and becomes something the enterprise must defend.



PREVENT

AI Policy Foundry

Generate & maintain cloud-security policy from live intel + the org's risk register.



PROTECT

AI-SPM

Govern the org's own AI/ML assets — adversarial defence, drift, compliance evidence.



DETECT & RESPOND

AI SOC Portal

Auditable, AI-native triage, hunting & response over live security telemetry.

◦ One trust spine across all three — every AI decision is explainable, evaluated and logged. Gaps found in Detect feed back to Prevent; assets in Protect are watched by Detect.



THE PAIN TODAY

- Defensive tooling is fragmented — policy, AI-asset posture and SOC detection are separate point tools that don't share context.
- AI is now both a defender and a new attack surface, yet nothing covers prevent → protect → detect as one governed, auditable loop.



THE INTEGRATED SOLUTION

- One platform spanning the lifecycle, sharing a common trust spine so every AI verdict is traceable, evaluated and audit-ready by design.
- Land in any one pillar; expand across the loop as the customer's AI footprint — and their obligation to govern it — grows.

The AI Defence Ecosystem — Why It Wins, and How We'll Know

The opportunity to exploit, what sets it apart, and the metrics behind it — modelled on the public builds; production figures under NDA.



THE OPPORTUNITY TO EXPLOIT

- AI-security spend is surging but fragmented into point tools — no one owns the whole lifecycle.
- Regulation (EU AI Act, NIST AI RMF) is forcing every enterprise to govern its AI — a mandatory, expanding buyer need.
- "AI as defender" and "AI as attack surface" are treated as one problem — a category few competitors frame this way.
- Land-and-expand: enter via one pillar, grow across the loop with the customer's AI footprint.



OUR DIFFERENTIATOR

- Full-lifecycle coverage — prevent → protect → detect & respond — vs. single-point competitors.
- Trust as a first-class product layer: every verdict traceable, evaluated and auditable, not a later hardening pass.
- Grounded in the customer's own risk register and AI assets — contextual, not generic templates.
- Governed, human-in-the-loop agentic action — never an autonomous black box.



SUCCESS METRICS

modelled on the public builds · production figures under NDA

~70%

● faster time-to-policy

vs. 2–6 week manual baseline

94%+

● policy & control coverage

up from 60–75% manual

100%

● AI verdicts auditable

traceable & replayable by design

3 → 1

● point tools consolidated

one governed platform loop

Exploit Validation & Remediation MCP Server

A governed MCP bridge that lets AI agents run the validate → approve → remediate loop

Implemented for: Qualys

 View on GitHub github.com/mayukhg/ai-agent-mcp

BUILT WITH

Python FastMCP · MCP resources / tools / prompts ·
stateful in-memory sandbox · Claude Desktop



CUSTOMER PAIN

Vulnerability management is a broken loop: an alert fires, but proving the CVE is really exploitable, getting approval, patching and re-verifying happens manually across tools — and no one dares let AI touch remediation without guardrails.



THE PRODUCT BET

An industry-standard MCP server so a conversational AI can safely drive a closed-loop validate→approve→remediate→re-verify workflow — human-in-the-loop gated. A new consumption channel that meets engineers in their chat/IDE.



WHY AI IS THE RIGHT CALL

AI owns the natural-language front door and orchestration — turning "verify this CVE and patch if real" into safe, stateful tool calls. Validation & remediation stay deterministic; approval stays human. AI orchestrates, it doesn't decide.



KEY TRADEOFF

Kept validation and remediation deterministic and gated approval to a human — deprioritised full autonomy so security teams can actually adopt it.



VALIDATED WITH

Built on direct customer feedback (Qualys) asking to stay flexible across their own and 3rd-party tooling — which drove the open, standard MCP-server design over a closed, one-off integration.



VALUE & OUTCOMES

- Extends the risk platform to safe AI agents
- Cuts mean-time-to-remediate
- Roadmap: live-API mode, deeper autonomy matrix

Unified Risk Copilot

One conversation that joins 8 security modules into a ranked, auditable remediation plan

Implemented for: Qualys

 [View on GitHub](https://github.com/mayukhg/unified-risk-copilot) github.com/mayukhg/unified-risk-copilot

BUILT WITH

qualys-cli-mcp wrapper · Node full-stack demo · module allow-listing · append-only JSONL audit



CUSTOMER PAIN

Teams run 8+ modules (VM, WAS, PC, TotalCloud, Container, Cloud Agent, Patch, CSAM) in separate UIs. There's no single "what to fix first" view, the CISO's weekly report is a manual spreadsheet, and managers want data without write access.



THE PRODUCT BET

A "Weekly Risk Copilot" on the official MCP server: an LLM joins all 8 modules into one ranked, evidence-backed narrative — with module allow-listing, read-only mode, audit logging and approval-gated remediation baked in.



WHY AI IS THE RIGHT CALL

Cross-domain correlation, NL querying and report synthesis are the LLM sweet spot — the join logic is fuzzy and the output is prose. Governance is deterministic guardrails; remediation is human-approved. Right split of AI vs. controls.



KEY TRADEOFF

Shipped read-only + approval-gated first — deprioritised write/automation to earn trust before ever letting AI act across 8 modules.



VALIDATED WITH

Same customer signal (Qualys) — the need to stay flexible across the stack — led to wrapping the official MCP server with module allow-listing, not building yet another siloed console.



VALUE & OUTCOMES

- Weekly risk report: hours → minutes
- Defensible, fully audited, safe self-service
- Roadmap: live qualys-cli, more agent actions

HS Code Prediction API

Automating customs classification from a plain-text product description

Implemented for: Pitney Bowes

 [View on GitHub `github.com/mayukhg/predict-hscode`](https://github.com/mayukhg/predict-hscode)

BUILT WITH

FastAPI · scikit-learn (TF-IDF + Logistic Regression) · Docker
· health checks · structured logs



CUSTOMER PAIN

Customs classification (assigning HS codes to products) is manual, slow and error-prone. A wrong code means fines, shipment delays and compliance risk — and trade teams still look codes up by hand, item by item.



THE PRODUCT BET

A production-shaped microservice that predicts an HS code — with a chapter/heading/subheading breakdown and confidence — from a free-text description. Drops into a trade/logistics pipeline as a simple, cheap-to-serve API.



WHY AI IS THE RIGHT CALL

A canonical supervised text-classification task — so it uses classic ML (TF-IDF + logistic regression), NOT an LLM. Lightweight, explainable and cheap to run: matching model complexity to the problem is itself the product decision.



KEY TRADEOFF

Shipped classic ML at ~55% rather than chase an LLM — a cheap, explainable, deployable pipeline beat SOTA accuracy for this buyer's need.



VALIDATED WITH

eBay's need to show an accurate total landed cost at checkout drove it; validated with customs domain experts on how HS codes are really assigned, which shaped the classify-from-text approach.



VALUE & OUTCOMES

- Accurate landed cost at checkout; less manual work
- Ship-ready, cheap-to-serve ML packaging
- Roadmap: transformer models, batch predict, caching

KYC-as-a-Service

Verifiable, privacy-preserving credit-risk scoring — AI on a blockchain

Implemented for: **Western Union**

 [View on GitHub `github.com/mayukhg/kyc-as-a-service`](https://github.com/mayukhg/kyc-as-a-service)

BUILT WITH

React + TypeScript · GBM (XGBoost/LightGBM) ·
permissioned blockchain · federated learning · OAuth2



CUSTOMER PAIN

Banks need real-time, auditable credit-risk scores, but AI scoring is a black box regulators distrust — and institutions can't pool data to improve models without exposing customer PII. Trust and privacy block AI adoption in lending.



THE PRODUCT BET

A three-layer hybrid platform: blockchain as immutable source-of-truth & model provenance, off-chain AI for scoring, and a signed API — plus federated learning so institutions improve the model without ever sharing raw data.



WHY AI IS THE RIGHT CALL

Credit scoring on tabular/behavioural features is a supervised ML problem, so it uses GBM, not an LLM. The real innovation is wrapping AI in verifiability (blockchain) and privacy (federated learning) — the barriers, not the maths.



KEY TRADEOFF

Invested in verifiability (blockchain) and privacy (federated learning) over model sophistication — those were the real barriers to adoption, not the GBM itself.



VALIDATED WITH

Western Union customers couldn't get credit in a new country without history or documents; validated KYC requirements with multiple financial institutions, which drove the verifiable, privacy-preserving design.



VALUE & OUTCOMES

- Adoptable AI scoring for regulated finance
- Multi-institution collaboration, GDPR-aware
- Solves thin-file cold-start via shared learning

Ellyra Pulse

Safety-gated NPS & sentiment intelligence for a medical AI companion

Implemented for: Ellyra Health

 [View on GitHub `github.com/mayukhg/ellyra-pulse`](https://github.com/mayukhg/ellyra-pulse)

BUILT WITH

React 19 + TypeScript · TanStack Start · PostgreSQL
· JWT + HMAC-signed ingest · PagerDuty webhook · Vitest



CUSTOMER PAIN

One aggregate NPS hides three very different failures — a UI annoyance, a cold tone, or an AI answer that broke someone's trust in their own care. Verbatims routinely carry PHI, and a genuine safety miss (a misread decimal on a blood panel) sits in a weekly report instead of reaching a clinician in minutes.



THE PRODUCT BET

A feedback pipeline where a clinical risk gate runs before any survey fires, PHI is redacted before storage or any model call, and aspect-level sentiment separates detractor volume from detractor severity — so safety signals page clinical on-call instead of becoming one more dashboard row.



WHY AI IS THE RIGHT CALL

Medical verbatims are unstructured and multi-aspect — “accurate, but felt cold” is two signals. ABSA across clinical trust, tone, OCR quality and actionability surfaces the rare, trust-breaking failures a theme-frequency count buries. Safety routing stays deterministic: models add evidence, never gate a P0.



KEY TRADEOFF

Chose an auditable, rule-based safety gate over a smarter LLM classifier, and shipped schema, routing and dashboard before deeper analysis (SHAP drivers, root-cause quadrant) — a clinician must be able to see exactly why a case was paged.



VALIDATED WITH

Ran the pain with Ellyra's clinical and product teams. An NPS score that can't tell a UI gripe from a broken-trust safety miss — and PHI in free-text feedback — is what drove the pre-survey risk gate and redact-before-storage design.



VALUE & OUTCOMES

- Safety-flagged detractors: 15-min SLA vs. 24h, with automatic clinical on-call paging
- Targets: anxiety relief >75%, comprehension >85%, inaccuracy flags <0.2%
- Roadmap: medical NER redaction, live UI-to-API wiring, SHAP driver analysis

PORTFOLIO SUMMARY

The Portfolio in One View

8 Products

3 Arenas

7 Domains

5 AI Techniques



Breadth of the Portfolio

- 8 products across 3 arenas — AI Defence, Agentic Platform Extensions, Applied AI.
- From cloud security & SOC to trade, customs, lending and healthcare.
- Every approach: multi-agent LLMs, RAG, MCP agents, classic ML, federated learning.



Business Value

- Cuts cost & risk — faster policy, fewer gaps, automated compliance evidence.
- Opens new channels — an agent-drivable platform and conversational self-service.
- Platform economics: land in one pillar, expand across the lifecycle.



User Value

- Meets users in their tools — chat, IDE, existing workflows — not one more dashboard.
- Hours of manual cross-referencing become minutes, audit-ready by default.
- Trust built in: every AI verdict is explainable, traceable and reversible.



Product Vision

- Start from the customer's real blocker, then ask whether AI genuinely helps.
- Right-size the tech — LLM, classic ML or governance, never AI for its own sake.
- AI as a governed teammate: human-in-the-loop, defensible, enterprise-ready.

HOW I BUILD AN AI PRODUCT — from ideation to business & user value

1



2



3



4



5



6

Ideate & Frame

Anchor on the customer's real pain.

Discover & Validate

Test the pain; size impact & fit.

Decide on AI

LLM, classic ML or governance — right-sized.

Architect & Build

Agents, RAG, MCP + a trust layer.

Demo & Iterate

Ship a zero-install demo; evaluate.

Deliver Value

Measurable business & user value.

The through-line: *relentless focus on the customer's problem — and the judgment to apply AI only where it earns its place.*

WORK WITH ME

A product leader for the **agentic-AI** era

Eight working AI products across enterprise security, trade, finance and healthcare — each one the same discipline: start from real customer pain, decide whether AI genuinely helps, and ship business and user value.



Range with depth

8 products, 3 arenas — cloud security & SOC to customs, lending and healthcare; every approach from multi-agent LLMs to classic ML.



Judgment over hype

Right-size the tech: LLMs where reasoning matters, classic ML where it's cheaper, governance where earning trust unlocks adoption.



Value-obsessed

Every product ties to business value (cost, risk, new channels) and user value (their own tools, hours-to-minutes, trust built in).



Ship, don't slideware

Every product has a working, zero-install demo — taken from ideation to delivered value through a repeatable process.

Seeking: Director / VP-level product leadership building AI or security products in the enterprise — where the hard problems are trust, adoption and turning capability into outcomes.

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